🏪 RS001 Week 4 — English Worksheet

Topic: Parallel Lists — A Shop Scene · **Course:** Text Adventure (Python) · **Time:** about 45 minutes

This week is about **parallel lists** — several lists that line up by index. Item `[1]`'s name, power, and price all sit at index `1` in three different lists. Together they build a shop. In this worksheet you will read code about parallel lists, think carefully, and then explain the shop code you wrote in today's live lesson.

```
shop_weapons = ["wooden stick", "iron sword", "magic staff"]
shop_power   = [2, 8, 15]
shop_price   = [5, 30, 80]
game.say(f"{shop_weapons[1]} — power {shop_power[1]}, {shop_price[1]} gold")
```

> 🧠 Words to know: **parallel list**, **index**, **power**, **price**, **line up**

1 · Predict 🔮

Read each piece of code. Run it in your head and write what you think the screen will show.

```
names = ["dagger", "axe", "spear"]
price = [10, 40, 25]
game.say(f"{names[1]} costs {price[1]} gold")
```

Which item and which price get printed together?

```
names = ["dagger", "axe", "spear"]
power = [3, 9, 6]
game.say(f"{names[0]} — power {power[2]}")
```

The indexes do not match. What strange sentence comes out?

```
names = ["dagger", "axe"]
price = [10, 40]
game.say(f"{names[2]} costs {price[2]} gold")
```

Each list only has 2 items. What happens when the code asks for `[2]`?

2 · Spot the Bug 🐛

Each block was meant to do something, but it is broken. Read what it is **supposed** to do, fix it on paper, then explain why the original was wrong.

Bug A — This shop line should show the **axe** with **its own** price. Right now it shows the axe's name but the dagger's price.

```
names = ["dagger", "axe", "spear"]
price = [10, 40, 25]
game.say(f"{names[1]} costs {price[0]} gold")
```

Hint: the name and the price should share the same index.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A new weapon `"hammer"` was added to the names list, but the other lists were not updated. Now line 4 crashes when it reaches the hammer.

```
names = ["stick", "sword", "staff", "hammer"]
power = [2, 8, 15]
price = [5, 30, 80]
game.say(f"{names[3]} — power {power[3]}, {price[3]} gold")
```

Hint: if one list grows, the parallel lists must grow too.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — This should print a full 3-line shop menu. Right now line 2 forgot to change its index, so the same weapon shows up twice.

```
names = ["stick", "sword", "staff"]
price = [5, 30, 80]
game.say(f"1. {names[0]} — {price[0]} gold")
game.say(f"2. {names[0]} — {price[1]} gold")
game.say(f"3. {names[2]} — {price[2]} gold")
```

Hint: look closely at the name index on line 2.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code 📖

Here is a working shop menu built from three parallel lists. Read it slowly, then answer the questions below.

```
names = ["stick", "sword", "staff"]
power = [2, 8, 15]
price = [5, 30, 80]
game.say(f"1. {names[0]} — power {power[0]}, {price[0]} gold")
game.say(f"2. {names[1]} — power {power[1]}, {price[1]} gold")
game.say(f"3. {names[2]} — power {power[2]}, {price[2]} gold")
```

Why are there three separate lists instead of one?

Look at line 5. Which name, power, and price get printed together, and how do you know?

The index goes 0, then 1, then 2 down the menu. What is that index doing for the player?

If you changed `names[1]` to `names[2]` on line 5, what would go wrong on the screen?

What would break if the `price` list had only two numbers in it?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote your own shop scene with parallel lists. Take a short video on your phone (or a parent's phone) explaining the code **you** wrote. You may show it running. Teach it like the viewer has never coded, and try to use these words: **parallel list**, **index**, **power**, **price**, **line up**.

> 1. Show your three lists and say what each one holds.
> 2. Read one weapon out loud — its name, power, and price.
> 3. Point to where all three sit at the same index.
> 4. Explain what would break if one of your lists were shorter than the others.

Write what you will say in your video. Plan it here before you record.

Submit ✅

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.